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REMANENT MAGNETISM IN SILTS OF PLEISTOCENE AGE FROM NORTH AMERICA[1]

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ABSTRACT

Remanent magnetizations of 64 silt specimens of Kansan, Illinoian, Sangamon, and Wisconsin age are reported. All, except some of the Illinoian specimens, are apparently normally magnetized. The primary magnetic fabrics of the Kansan, Sangamon, and Wisconsin specimens may have been deformed slightly by the consolidating effects of overriding Pleistocene ice sheets. Twenty-two specimens from the basal part of an Illinoian glaciolacustrine silt display a confusion of north-magnetic-pole plots, 21 of the plots lying between latitudes 18°S to 88°N and longitudes 7° to 176°W. This apparent confusion of the magnetic field corresponds to the time of the Illinoian glacial maximum in southern Indiana. Nine specimens from the upper part of the Illinoian silt unit, representative of post glacial-maximum time, are normally magnetized.

INTRODUCTION

Irving's (1959, 1960) comprehensive survey and analysis of paleomagnetic pole positions indicates that most of the Pleistocene positions that have been determined are assignable to either early or very-late Pleistocene time. A few pole positions are known for "middle-Pleistocene" time, but the precise moments within the epoch to which the pole positions may be assigned are unknown. In this report are presented the results of an investigation into the remanent magnetizations of silt samples of middle to late Pleistocene age from well-understood stratigraphic sections of North America.

In 1958 and 1959, Harrison collected oriented samples for magnetic analysis from Indiana and Illinois; Terasmae and P. F. Karrow collected samples from Ontario and Quebec. Magnetic analysis was performed by J. Mead, using a "spinner"-type magnetometer, while specimen preparation and data calculations were accomplished by Harrison. The authors are indebted to Dr. Mead for making the magnetic measurements.

LOCATIONS, STRATIGRAPHY, AND SEDIMENTATION

Sixty-four specimens, representing small time intervals of the Kansan, Illinoian, Sangamon, and Wisconsin stages of the Pleistocene epoch were studied. Magnetic pole positions (table 1) were calculated by averaging the magnetic vectors of each group of specimens and plotting the mean vector according to the methods described by Graham (1955). Where the dispersion of the magnetic inclinations for a group of closely associated specimens was such that the semi-axis, $d\chi$, of the elliptical error area around the pole (at a probability of 95 percent) was less than 14, the vectors were not averaged; instead, the individual pole positions, as determined from the magnetic vectors for each specimen, were listed individually (table 1, col. 6).

Kansan Samples

The oldest of the Kansan samples were cut from a bed of silty alluvium in western Illinois. Horberg (1956) considered the alluvium to be early Kansan in age. The bed of noncalcareous, unoxidized silty clay exhibited occasional layers of well-preserved twigs of conifers; it cropped out 6 in above stream level near 39° 52′ 30″ N. Lat., 91° 18′ W. Long. Because it was overridden by the Kansan and Illinoian ice sheets, the bed is not normally consolidated but overconsolidated.

Samples of a proglacial silt, considered by Wayne (1958, p. 10) to be of Kansan age, were taken in west-central Indiana (near 39° 28′ 30″ N. Lat., 86° 54′ 30″ W. Long.). The samples were cut from the laminated lake

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TABLE 1.—*Magnetic-north-pole positions obtained from remanent magnetizations of specimens of silts of Pleistocene age from North America*

Footnotes are to studies that give stratigraphy and (or) sedimentation at sampling localities.

Stratigraphic Position (1)	No. Loc. (2)	No. Spec. (3)	Pole Positions (4) Lat.	Long.		(5) Spec. No.	(6) Pole Positions Lat.	Long.
Wisconsin:								
Glacial maximum[1]	1	5	56N	138E				
Scarborough beds[2]	2	11	73N	35E				
Sangamon:								
Upper Don beds[3]	1	2	69N	175E	Vertical Samplin Profile	22	18S	86W
						21	12N	148W
Middle Don beds[3]	1	2				20	20N	145W
						19	8S	74W
Basal Don beds[3]	1	5	84N	170W		18	1N	176W
						17	48N	96W
Illinoian:						16	4N	152W
						15	50N	145W
Post glacial maximum[4]	1	9	71N	33W		14	4N	7W
						13	88N	156W
Glacial maximum[5]	2	22	31N	112W		12	35N	81W
						11	80N	88W
						10	0	77W
Kansan:						9	80N	150E
						8	24N	71W
Preglacial maximum[6]	2	6	64N	57E		7	61N	141W
						6	4N	84W
Early alluvium[7]	1	2	80N	65E		5	45N	156W
						4	20N	42W
						3	36N	107W
						2	46N	41W
						1	31N	173W

[1] Harrison, 1958, table 1, no. 6, and appendix, no. 6; also, Harrison, in preparation.
[2] Terasmae, unpublished studies; (specimens from basal 1.0 meter).
[3] Coleman, 1914, 1933.
[4] Wayne, 1958, p. 12, units 7 and 8; (3 specimens taken along one bed are in agreement).
[5] Wayne, 1958, p. 12, unit 4, (basal 0.6 meter).
[6] Wayne, 1958, p. 10, unit 3; (3 specimens taken along one bed are in agreement).
[7] Horberg, 1956, appendix, no. 13; (change location to NW¼ SE¼ of section).

silt described by Wayne (1958, unit 3); this bed is overconsolidated, owing to the stresses of overriding Kansan and Illinoian ice sheets.

Illinoian Samples

Several samples of a laminated glaciolacustrine silt of Illinoian age were taken near 39° 16′ 30″ N. Lat. and 86° 25′ W. Long. in south-central Indiana. Petrography of the silts is given by Smith (1956) and the stratigraphy by Wayne (1958, p. 12). Specimen cubes cut from samples from the middle zone of the silts were too weakly magnetic to permit accurate measurements, and so the magnetic data of table 1 are presented in two groups, representing the upper and lower samples. The valley in which the lake silts were deposited was dammed upstream and downstream by southward moving ice of Illinoian age. Assuming Flint's (1955) rates of overland ice advance and retreat, the lake may have existed 2500 to 4500 years while the ice advanced to and retreated from the Illinoian drift boundary. It is possible, therefore, that the lowermost

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Illinoian silts were deposited about the time of the glacial maximum (table 1, col. 1), but that the uppermost silts were deposited during glacial retreat, some 2000–4000 years later. The uppermost samples of the lake silts are overlain by only 3 ft or so of laminated deposits and then by 6 ft of outwash sands and gravels that attest, probably, to *rapid* melting of the retreating Illinoian ice. Finally, 17 ft of noncalcareous silt, that is probably mostly loess, overlies the outwash sands and extends to the upland surface.

All of the cubes for magnetic analysis were cut from the more clayey layers of the laminated beds, where the beds were calcareous, unoxidized, and undesiccated. Where sampled, the beds were composed of about 30 to 40 percent clay (mainly illite and chlorite) and 55 to 65 percent very-fine silt. Smith (1956, nos. 18–27) reported that a series of channel samples through the silts had an average *p*H of 8.0 and contained detrital magnetite and organic material in the $>49\mu$ fraction (amounting to less than about 2 percent of his samples). The greatest dips observed in the undisturbed, laminated beds were 3° to the northeast. The silts were undoubtedly in a state of normal consolidation and were sampled in the undisturbed state.

There are many similarities between the glaciolacustrine silts of Illinoian age of this study and the glaciolacustrine silts of Wisconsin age from New England that were studied by Johnson, Murphy, and Torreson (1948). Both the Illinoian and the Wisconsin silts are laminated, physically and chemically undisturbed, and normally consolidated. The magnetic vectors found in the specimens of the basal Illinoian silts (table 1), however, are much more steeply inclined than are the vectors that were found by Johnson and others in the Wisconsin silt specimens. The nine specimens from near the top of the Illinoian lake clays, on the other hand, have inclinations (table 1) comparable to those found in the Wisconsin silt specimens from New England.

Sangamon Samples

Silt beds of Sangamon age (table 1) were sampled at three places around the periphery of the Don Valley Brickyard pit near Toronto, Ontario (43° 40′ N. Lat., 79° 22′ W. Long.). Coleman's studies (1914, 1933) of the macroscopic flora and fauna, and Terasmae's study (1958) of the pollen spectra from the Don Beds indicate that a portion of these beds was deposited when the mean annual temperature was probably 3–5° C warmer than the present at Toronto. The beds are an alternating series of alluvial and lacustrine sands, silts, and gravels. They have been consolidated to an unknown degree as a result of stress induced by the thick sheet of overriding Wisconsin ice. Cubes for magnetic analysis were cut from samples of calcareous, unoxidized silt beds.

Wisconsin Samples

The oldest samples of Wisconsin age (table 1) were obtained from the Scarborough beds at the Don Valley Brickyard pit at Toronto, Ontario. The Scarborough beds, where sampled, were lacustrine silts and were calcareous, unoxidized, and undisturbed. They were consolidated by the overriding Wisconsin ice sheet to the same extent, probably, as the underlying Don beds of Sangamon age.

An interstadial silt unit of possible early-Wisconsin age (Harrison, in preparation), dated at >38,000 B.P. (W-814, W-578), was sampled near 39° 52′ 30″ N. Lat. and 86° 18′ W. Long. Where sampled, the silt beds were calcareous, unoxidized, horizontally laminated, and undisturbed. The silt unit has been consolidated (Harrison, 1958, tab. 1, no. 6) to a probable value of 43.0 kg/cm² (to a minimum value of 26.0 kg/cm²); this amounts to an *overconsolidation* of, probably, 38.4 kg/cm², the present effective overburden pressure being 4.6 kg/cm².

REMANENT MAGNETISM

The remanent magnetism ranged in intensity from approximately 2×10^{-5} cgs unit per cc in some of the specimens of Illinoian age to 45×10^{-5} cgs unit per cc in some specimens from the Don beds. A *primary* magnetic fabric undoubtedly was imparted to the silts at the time of their formation, when detrital ferromagnetic particles became oriented in the earth's magnetic field as they settled out of aqueous suspension.

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POSSIBLE MODIFICATIONS OF THE PRIMARY MAGNETIC FABRIC

Irving (1959) has considered the possibilities whereby sediments fail to preserve the direction of the geomagnetic field imparted to them at the time of their formation. He indicates that little is known of the effects upon the primary magnetic vector produced by post-depositional overconsolidation of detrital sediments. All but the Illinoian sediments of this study are overconsolidated to varying degrees; their in-place void ratios are many times less than those which would be expected if they had been consolidated only by the sediments that presently overlie them.

Consider the silt unit deposited about the time of the early Wisconsin glacial maximum (table 1). The north-magnetic-pole position (table 1, col. 4) obtained for the silt unit lies a few degrees south of the limit of the expectable secular variation (approximately the 60th parallel) and in the *eastern* hemisphere. Because the bed has been overconsolidated by about 38.4 kg/cm², one might expect that the average inclination of the primary magnetic fabric has been "flattened" toward the horizontal plane during sediment consolidation by overriding Wisconsin ice some 425 m (1400 ft) thick. One can infer that silt rebound (Harrison, 1958, no. 6), since overconsolidation by now-vanished ice, has not been of sufficient magnitude to permit "relaxation" to the degree required for restoration of the primary magnetic fabric. It is probable that all of the other specimens (except the Illinoian) have been overconsolidated to an even greater extent than the "Wisconsin maximum" samples. Because of this fact, there is some difficulty in relating the magnetic pole positions found for these samples to "absolute" geographic pole positions.

The following information may be cited in support of this view. Four silt specimens were obtained from a bed of normally-consolidated Pleistocene lake silts in western New Hampshire (43° 41′ 30″ N. Lat., 72° 16′ 49″ W. Long.). The bed occurs in an exposure of "varved clays" of the same type and age as those studied by Johnson and others (1948). One specimen from the bed was stressed in the soil-mechanics laboratory at Dartmouth College to a value of 160 kg/cm². This is equivalent to roughly 1,775 m (5825 ft) of overriding glacial ice. A specimen cube the size of one of the three unstressed specimens was then made up by combining two thicknesses of the stressed specimen in identical horizontal orientation. The results indicated that the magnetic vector of the specimen stressed in the laboratory had been flattened about 10° toward the horizontal when compared with the vectors of the unstressed specimens.

PALEOMAGNETISM OF ILLINOIAN SPECIMENS

The normally-consolidated specimens of Illinoian silts (table 1) are thought likely to exhibit primary magnetic fabrics. Detailed studies by Johnson and others (1948), of very similar sediments of Wisconsin age indicate that laminated, normally-consolidated, glaciolacustrine silts are among the best of the clastic sediments to measure for purposes of paleomagnetic reconstructions. There is at present no reason to suspect that the magnetic vectors of the Illinoian specimens are not representative of primary magnetic fabrics.

Assuming that the geomagnetic field throughout geologic time tends to be symmetrical about the earth's axis of rotation, one is tempted to infer shift of the geographic north pole from the paleomagnetic data of these Illinoian specimens. The wide dispersion of the magnetic pole positions (table 1, col. 6) for the 22 specimens must be satisfactorily explained, however, before shift of the geographic pole can be postulated. The wide dispersion might possibly reflect the *disturbance* of the geomagnetic field during a slight shift of the earth's axis of rotation.

Momose's (1958) measurements of Japanese volcanic rocks and lacustrine sediments indicate continuous shift of the centered dipole from Lat. 75° N. to Lat. 70° S. during Pliocene times. Momose's rate of magnetic polar shift is considerably less than that implied by our two groups of Illinioan specimens (table 1, cols. 1, 4). Theoretically, however, the rate of pole migration could be much greater (Gold, 1955, p. 528). The effect of rapid geographic polar shift on

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the geomagnetic field, unfortunately, is indeterminable.

SUMMARY AND CONCLUSION

Measurements of the remanent magnetizations of silt specimens of Kansan, Sangamon, and early Wisconsin age indicate magnetic-north-pole positions within the expectable secular variation (north of the 60th parallel, approximately), but these pole positions may not be the true positions during the time of silt deposition. A secondary magnetic fabric may have been imparted to the silts as a result of the stresses exerted upon them by overriding Pleistocene ice masses followed by only partial intergranular readjustment after unloading by the ice.

Some evidence is given for movement of the centered geomagnetic dipole during the time of the Illinoian glacial maximum in Indiana. This evidence is contrary to other data (Irving 1959, 1960) that indicate considerable stability of the geomagnetic field throughout the Pleistocene epoch.

It is felt that additional measurements of the remanent magnetization of clastic sediments of Pleistocene glacial and interglacial age will prove profitable. Wherever possible, the sampled sediments should be in a state of normal consolidation and should occur in geologic sections that are well-understood stratigraphically.

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